

Samarth Dumasia

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Professional Summary

Motivated Computer Science Engineering student with hands-on expertise in ROS2, robotics software architectures, and hardware-software integration. Completed the specialized RRP Robotics program featuring the design and development of T-bot, an autonomous mapping and navigation robot using simulation and sensor integration. Experienced team leader with a strong track record of securing innovation grants.

Education

Bachelor of Technology (B.Tech.) in Computer Science Engineering

P.P. Savani University, Surat, Gujarat, India

CGPA: 7.47/10

Graduation: 2026

Technical Skills

Robotics & Simulation: ROS2, Gazebo, RViz2, URDF Modeling, SLAM

Hardware & Embedded: Arduino, NodeMCU, Raspberry Pi, Microcontrollers, Sensors, Actuators

Programming Languages: C/C++, Python, JavaScript, HTML, CSS

Development Tools: Arduino IoT Cloud, Git, GitHub, Android Studio, Figma

Specializations: Robotics, ROS2 Development, Hardware Prototyping, Bill of Materials (BOM) Management

Projects

T-bot – Autonomous Mapping and Navigation Robot

ROS2, Gazebo, RViz2, URDF, SLAM

- Designed and developed an autonomous mapping and navigation robot as part of the RRP Robotics Program.
- Implemented ROS2 nodes, sensor fusion, and navigation pipelines.
- Utilized Gazebo and RViz2 to validate motion planning and control algorithms.

Eyes On Wheels – IoT Drowsiness Detection System

IoT, Sensors, Embedded Systems, Machine Learning

- Designed and led end-to-end hardware-software development of an IoT-based driver safety solution.
- Managed component sourcing, prototyping, wiring, and hardware testing.
- Built reliable data pipelines and alert mechanisms.
- Secured an INR 35,000 grant under the Student Startup and Innovation Policy (SSIP).

Mute's Speaker – Sign Language Translation System

IoT, Gesture-to-Text, Machine Learning

- Engineered an assistive technology system for real-time sign language recognition.
- Integrated gesture sensors with processing units to generate text-to-speech outputs.
- Improved communication accessibility for individuals with speech disabilities.

Smart Home Monitoring System

Arduino IoT Cloud, MQTT, Embedded Systems

- Developed a home automation system integrating gas/smoke sensors, actuators, and microcontrollers.
- Implemented MQTT and Arduino IoT Cloud connectivity.
- Enabled Alexa and Google Assistant integration.
- Performed circuit planning, component selection, integration, and testing.

Academic Experience

IoT Tech Lead

Google Developer Group (GDG), P.P. Savani University

2024–2025

- Led and mentored 10+ students in embedded systems and IoT development.
- Organized technical workshops on sensor integration, automation, and microcontroller programming.

Project Team Leader

Eyes On Wheels Project

Surat, Gujarat

- Directed a cross-functional team of five members.
- Managed hardware assembly, procurement timelines, and resource allocation.

Active Member

Entrepreneur Business Club (EBC), P.P. Savani University

- Contributed to startup projects focused on technology innovation and hardware-software integration.

Certifications & Achievements

- Google Cloud Platform (GCP) Certification – Verified badges on Credly
- Amazon Web Services (AWS) Certification – Verified badges on Credly
- IoT (Internet of Things): Wireless and Cloud Computing – Yonsei University (Coursera)
- Python for Everybody – University of Michigan (Coursera)
- Work Smarter, Not Harder: Time Management for Personal and Professional Productivity – Coursera
- Student Startup and Innovation Policy (SSIP) Grant Recipient – INR 35,000 (2024–2025)

Additional Skills

Soft Skills: Leadership, Team Collaboration, Public Speaking, Project Management, Problem Solving, Fast Learning, Communication, Mentoring

Design Skills: UI/UX Design, Graphic Design, Video Editing, Web Design, Photoshop, Figma, Blender

Interests: Technology Innovation, IoT, Robotics & Automation, Professional Networking, Problem Solving Projects